MIL-C-20061D
25 June 1979
SUPERSEDING
MIL-C-20061C(SHIPS)
25 June 1960
(See 6.4)

MILITARY SPECIFICATION

CLINOMETERS, SHIP, AND ACCESSORIES

This specification is approved for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the bubble-in-tube type ship clinometers used for measuring the heel and trim angle of a ship.

1.2 Classification.

1.2.1 Clinometers shall be of the following types, as specified (see 6.1.1):

Type I - Trim clinometer.
Type II - Heel clinometer.

1.2.2 Clinometer tubes shall be of the following classes, as specified (see 6.1.1):

Class A - 15 degrees of arc each side of the vertical.
Class B - 20 degrees of arc each side of the vertical.
Class C - 60 degrees of arc each side of the vertical.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-P-206 - Felt, Sheet, Cloth Wool, Pressed.
O-E-760 - Ethyl, Alcohol (Ethanol) Denatured Alcohol, Proprietary Solvents and Special Industrial Solvents.
O-M-232 - Methanol (Methyl Alcohol).
DD-G-541 - Glass (Laboratory).
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-795 - Cushioning Material, Flexible, Cellular Plastic Film For Packaging Application.

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-S-901 - Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipments and Systems, Requirements For.
MIL-F-2312 - Felt, Hair or Wool: Mildew Resistant and Moisture Resistant Treatment For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-L-10547 - Liners, Case and Sheet, Overwrap: Water-Vaporproof or Waterproof, Flexible.
MIL-P-15024 - Plates, Tags and Bands, Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-P-15280 - Plastic Material, Unicellular (Sheets and Tubes).
MIL-I-16562 - Insulation, Synthetic, Rubber-like Chemically Expanded, Cellular (Sheet Form).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS36079 - Glass Tubing, Round (Standard Wall).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1 General. Clinometer shall consist of one or two clinometer tubes fastened by means of clips to a mounting board on which degree-graduated scales are located above each tube recess. Unless otherwise specified (see 6.1.1), clinometers shall be furnished complete with tubes.

3.1.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Mounting board. Mounting boards shall be fabricated of plastic material conforming to type MFI of MIL-M-14. Design of the mounting board shall be as shown on figures 1 and 2. Tubes shall be recessed in the boards to approximately one half the tube diameter. Boards shall be free from warpage and chipped or blistered surfaces. There shall be no burrs or projecting ridges at the dividing lines of the mold nor any defects in finishes at these points caused by chipping, filing, or grinding without subsequent buffing or polishing. Boards shall be finished a dull black. Boards for the type I clinometer shall provide for mounting one class A clinometer tube. Boards for the type II clinometer shall provide for mounting one class B and one class C clinometer tube.

3.2.1 Graduations. A graduated anodized aluminum plate conforming to type H of MIL-P-15024 and designed for normal service in accordance with MIL-P-15024/5 shall be furnished for use with the class A, B, and C clinometer tube. Design of the graduated plates shall be as shown on figure 3. The zero degree mark shall be at the center and the remainder of the degree markings equally spaced to the left and right of zero in one-degree increments. Each 5-degree mark shall be numbered. The length of the single degree marks shall be 1/4 inch and the length of the 5-degree marks shall be 5/16 inch. The graduations shall be 0.020 inch wide. The numerals shall be 1/4 inch high with a line width of 0.031 inch. The base of the numerals shall be located 1/16 inch from the top of the 5-degree graduations. The base of the graduations shall terminate at the bottom edge of the plate. The background of the graduated plates shall be a lusterless black with natural anodized graduations and numerals. The graduated plates shall be bonded in place on the mounting board using a general purposes synthetic base cement.

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3.3 Tube clips. Tubes shall be secured to the mounting board by means of brass clips shown on figure 4. Clips shall be fastened to the mounting board below the tube recess by means of a brass screw. Four clips shall be spaced equi-distant along the length of the tube with the two outermost clips located just beyond the extremities of the graduated scale. An additional clip shall be located midway on each shank on the tube goose neck. Clips and screws shall be finished a dull black.

3.4 Tube cushion. Tubes shall be cushioned by padding the tube recess in the mounting board with a 3/32 inch thick white felt or foam. Felt shall conform to type III, class 11A1 of C-F-206 and shall be impregnated in accordance with treatments specified in MIL-T-2312. Foam shall conform to type I of MIL-P-15280 or MIL-I-16562. Padding shall extend along the entire length of the tube recess and be entirely contained within the tube recess.

3.5 Tubes.

3.5.1 Material. Tubes shall be made of glass tubing conforming to type I, class A of DD-G-541. The tubes shall be clear and free from bubbles, pits, scratches, cracks or breaks, and other imperfections.

3.5.2 Diameter. The tube diameter and wall thickness shall be in accordance with part number MS36079-4 of MS36079.

3.5.3 Design. Design of the tubes shall be as shown on figures 5, 6, and 7.

3.5.4 Appearance. Tubes shall be of uniform curvature, of correct radius, and the axis of the tube shall be in the plane of the radius.

3.5.5 Air chamber. One end of the tube shall be provided with an air chamber for the purpose of regulating the size of the bubble. This air chamber shall be of the same diameter and wall thickness as the remainder of the tube, and shall have the same tolerances. When sealing the tubing, an air space equal to one-half the volume of the air chamber shall be allowed above the surface of the liquid when the atmospheric temperature is 21.1 degrees Celsius (°C) plus or minus 15.0 °C (see 4.6.2).

3.5.6 Spirits. Tube shall be filled with a solution obtained from a mixture of 5 parts of methyl alcohol as specified for Grade C of O-M-232 to 100 parts of denatured alcohol in accordance with type III of O-E-760. The mixture shall be dyed red by the addition of one part by weight of fuchsin to 500,000 parts by weight of the alcohol solution.

3.5.7 Repair parts. Clinometer tubes may be ordered separately under this specification as repair parts (see 6.1.1).

3.6 Accuracy. Any angular movement of the tube in a vertical plane shall produce a corresponding movement of the bubble within 1 degree for the type I, class A, and type II, class B tube, and within 2 degrees for the type II, class C tube (see 4.6.1).

3.7 Shock. The assembled unit shall withstand, without breakage of board or tube(s), Grade B, Class I, Type A, shock test in accordance with MIL-S-901 (see 4.6.3).

3.8 Identification plate. Anodized aluminum identification plates, conforming to type H of MIL-P-15024, shall be fastened to the mounting boards. Plate size shall be as shown on figures 1 and 2, and shall contain the following information:

(a) SHIP CLINOMETER
(b) TYPE
(c) CONTRACTING AGENCY
(d) CONTRACT NO. YEAR
(e) MANUFACTURER'S NAME
(f) NSN
(g) U.S

3.9 Workmanship. The workmanship on the clinometer boards shall be free from warpage, chipped or blistered surface, burrs, projecting ridges; finish shall not be chipped, filed, or grinded without buffing or polishing; the tube shall be clear, free of bubbles, pits, scratches, and other imperfections.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of test reports. Reports covering the inspection and tests for First article inspection and the Quality conformance inspections shall be furnished to the contracting activity when specified (see 6.1.2).

4.1.2 Material inspection. When specified (see 6.1.2), material inspection shall consist of certification supported by verifying data that materials used in the fabrication of the clinometers and accessories are in accordance with the requirements of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 and 4.6 and shall be conducted on one sample of each type and class.

4.4 Quality conformance inspection. Quality conformance inspections shall consist of the examination of 4.5 and the accuracy and air chamber test of 4.6.1 and 4.6.2, respectively.

4.4.1 Sampling.

4.4.1.1 Lot. Clinometers (or tubes) of the same type and class offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.4.1.2 Sampling for examination. Sample clinometers or tubes shall be selected at random from each lot in accordance with MIL-STD-105 at inspection level III and shall be subjected to the examination of 4.5. Any clinometer having one or more defects shall be rejected.

4.4.1.3 Sampling for accuracy and air chamber test. Sample clinometers or tubes shall be selected from each lot in accordance with inspection level S-4 of the table for small sample inspection of MIL-STD-105 for the tests of 4.6.1 and 4.6.2. If any unit fails at any point, the lot represented shall be rejected.

4.5 Visual and dimensional examination. The clinometer selected in accordance with 4.4.1.2 shall be visually and dimensionally examined to verify conformance with this specification. The curvature at either the top or bottom of the tube shall be compared with a template of the specified curvature. Any clinometer containing one or more defects shall be rejected. Defects are as follows:

(a) Mounting board not of type specified; clinometer tubes not of class specified.
(b) Clinometer mounting board not in accordance with figure; board not recessed as specified; no holes for mounting.
(c) Clinometer incomplete; parts missing.
(d) Tube mounting recess in mounting boards not required depth.
(e) Tube graduations not as specified.
(f) Tubes not secured to board with brass clips; clips not formed to curvature of tube; clips not positioned as required; not installed with brass screws.
(g) Tube not cushioned in mounting board recess with white felt or foam; felt not impregnated as specified; foam does not meet specification requirements; tube not padded entire length.
(h) Tubes not clear glass specified. Not free of bubbles, pits, scratches, cracks, breaks, or other imperfections.
(i) Tubes not of specified design for class of tube; not uniform curvature; not correct radius; axis not in plane of radius; radius not of specified length.
(j) Tube length not as required; outside diameter, wall thickness not within tolerance.

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(k) Air chamber not required length; bend not within allowable diameters.
(l) Tube not filled with specified alcohol mixture; alcohol not dyed red; tube not sealed; air space not required volume.
(m) Top or bottom of tube nonconforming to template.
(n) Tubes tipped off at wrong end.
(o) Identification plates not as specified.

4.6 Test procedures.

4.6.1 Accuracy. The clinometer shall be mounted so that it can be revolved in a vertical plane on a calibrated stand. The indications shall be observed when the clinometer is displaced from the vertical. Five points on each side of the vertical shall be tested to determine compliance with 3.6. This test shall be performed at a temperature between 18.3°C and 23.9°C.

4.6.2 Air chamber. The tube shall be turned so that the air chamber is filled with alcohol and the remainder of the alcohol is in one end of the tube. The amount of air and of alcohol shall be estimated by measuring the unfilled portion of the tube to determine compliance with 3.5.5.

4.6.3 Shock. The assembled unit shall be shock tested to determine conformance with 3.7. Units which have been subjected to shock test shall not be offered for lot acceptance without specific written approval of Command or Agency concerned.

4.7 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with MIL-P-116 to verify conformance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.2).

5.1 Packaging levels A and C. Package each clinometer, or tube, in an individual fiberboard box conforming to PPP-B-636, Type CF, Class weather resistant with minimum bursting strength of 200 pounds per square inch. The clinometer, or tube, shall be cushioned within the container with cushioning material conforming to Class I 1/2-inch cell size of PPP-C-795, to prevent damage from movement or shock. Minimum cushioning shall be 1 inch top, bottom, sides and ends. Each unit package shall be capable of passing the free fall drop test of MIL-P-116, with interpretation of results for this test as specified in MIL-P-116.

5.2 Packing. Packing shall be level A, B, or C (see 6.1.1).

5.2.1 Level A. Clinometer, or tube, preserved and packaged as specified in 5.1, shall be packed in overseas type wood-cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, wood-cleated, paper-overlaid, or wood-cleated plywood boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636 Class weather resistant, PPP-B-576, or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have caselliners conforming to MIL-L-10547, and shall be closed and sealed in accordance with the appendix thereto with Method V closure applicable to PPP-B-636 boxes. Caselliners for PPP-B-636 boxes are not required. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.2.2 Level B. Clinometers, or tubes, preserved and packaged as specified in 5.1, shall be packed in domestic type wood-cleated fiberboard, nailed wood, wirebound wood, cleated plywood or wood-cleated paper-overlaid boxes or class domestic fiberboard boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, PPP-B-576, or PPP-B-636, respectively, at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C. Clinometers, preserved and packaged as specified in 5.1, shall be packed in containers of the type size and kind commonly used for the purpose in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

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5.3 Marking. In addition to any special marking required by the contract or herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each package (see 5.1) shall be marked as follows:

"THIS PACKAGE SHALL BE OVERPACKED PRIOR TO SHIPMENT".

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of clinometer required (see 1.2).
(c) Class and quantity of clinometer tubes required, if ordered separately as repair parts (see 1.2.2 and 3.5.7).
(d) When tubes are not required (see 3.1).
(e) Level of packing required (see 5.2).

6.1.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data Requirements	Applicable DID	Option
4.1.1	First article and quality conformance inspection reports	UDI-T-23473	
4.1.2	Material certification	UDI-T-23741	

(Copies of data item description required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.1.2.1 The data requirements of 6.1.2 and any task in section 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

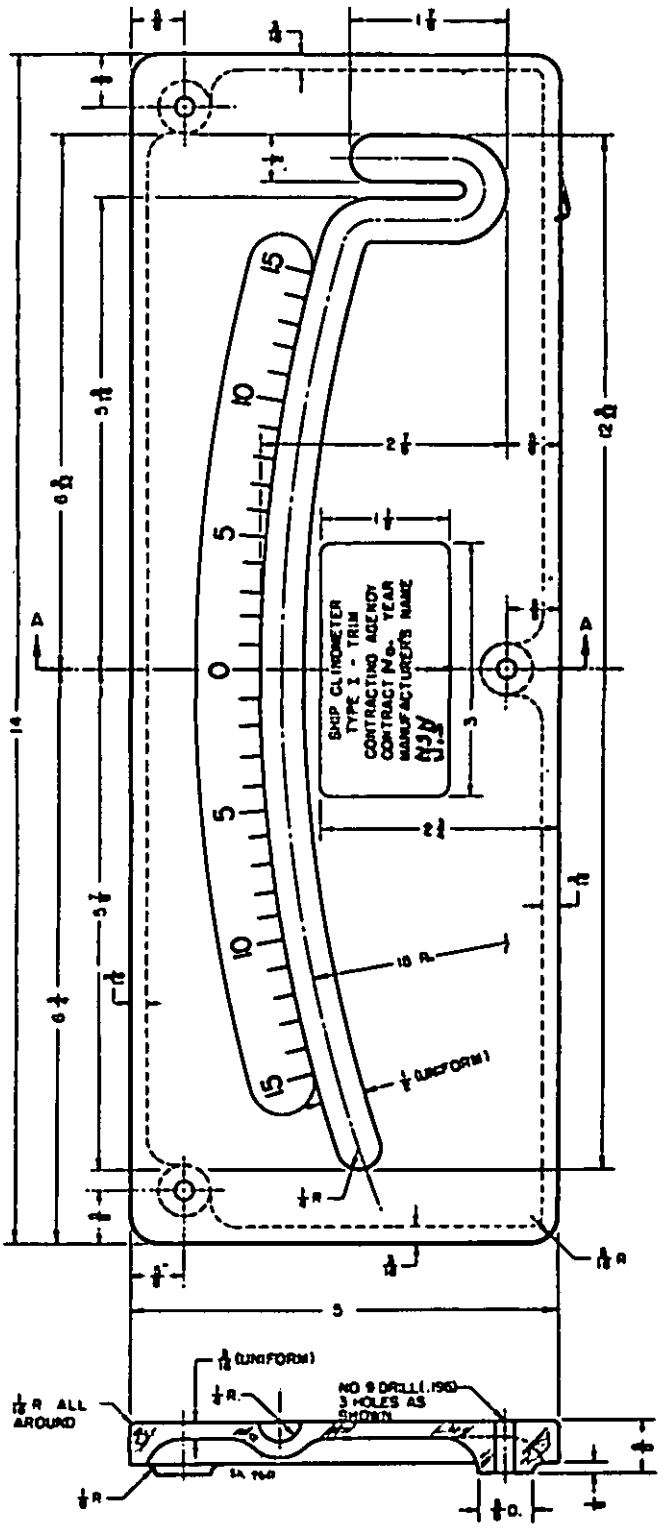
6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Changes from previous issue. "*" are not used in the revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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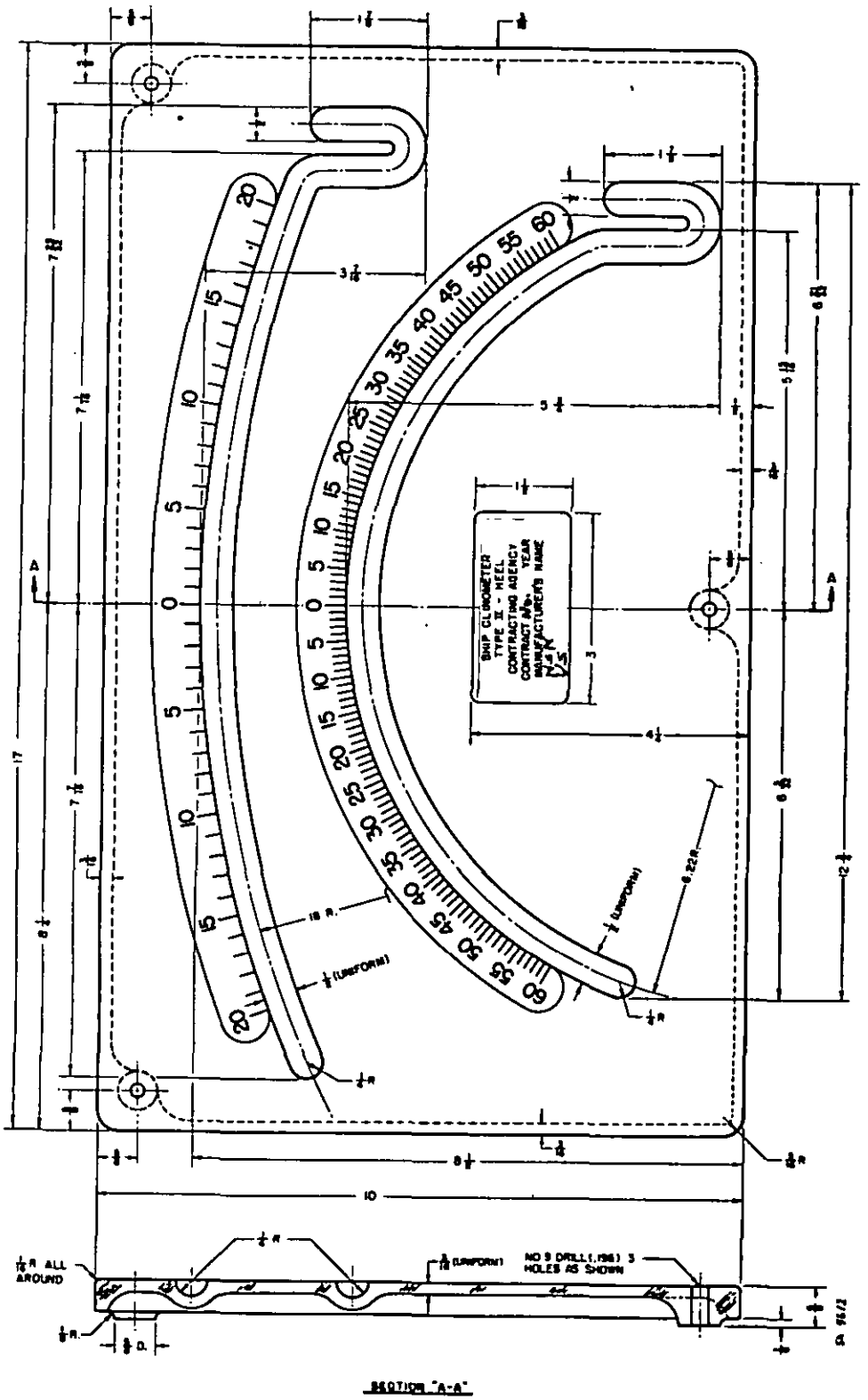
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NOTE: All dimensions are in inches.

FIGURE 1. Type I clinometer.

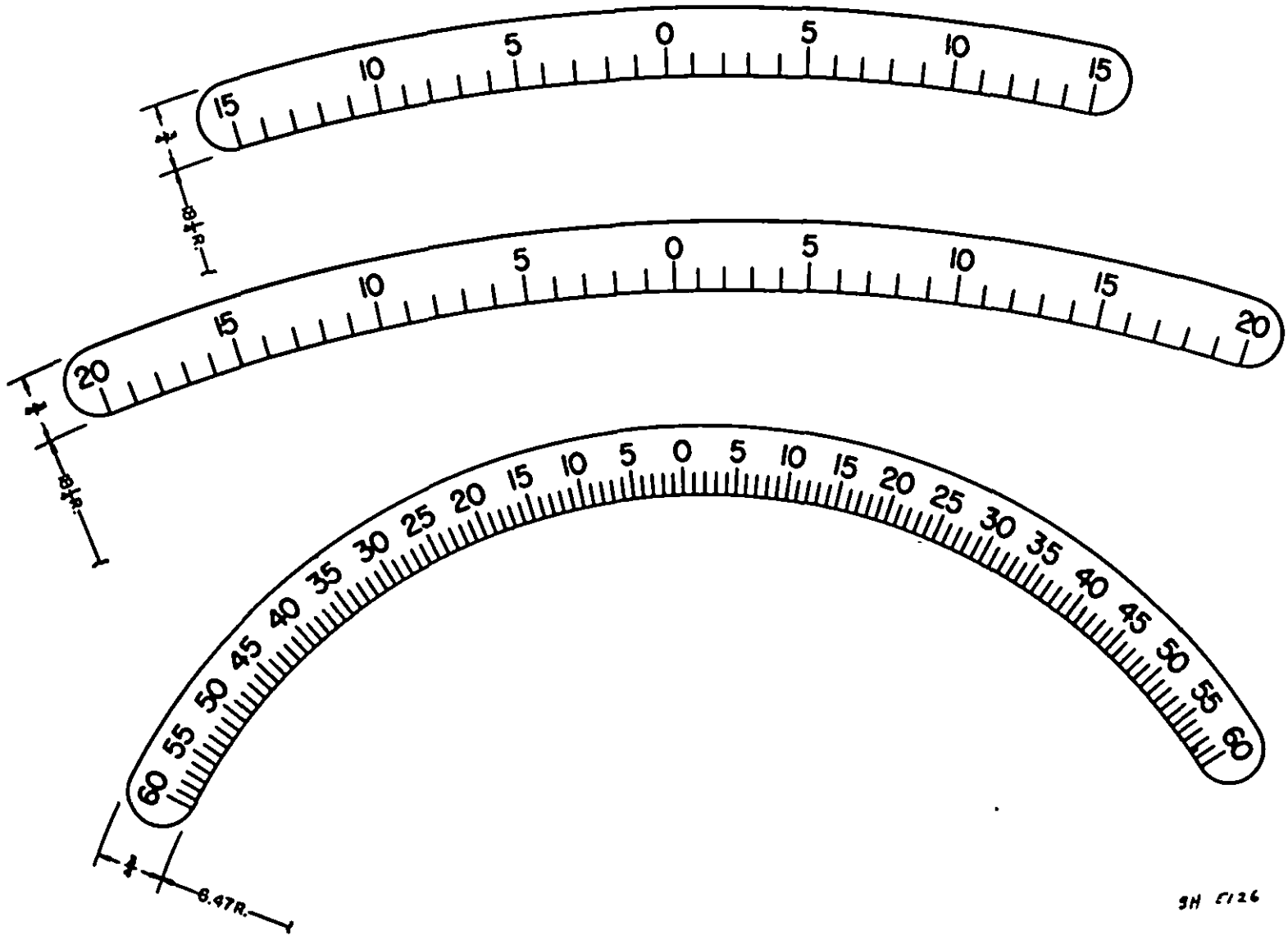
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FIGURE 2. Type II clinometer.

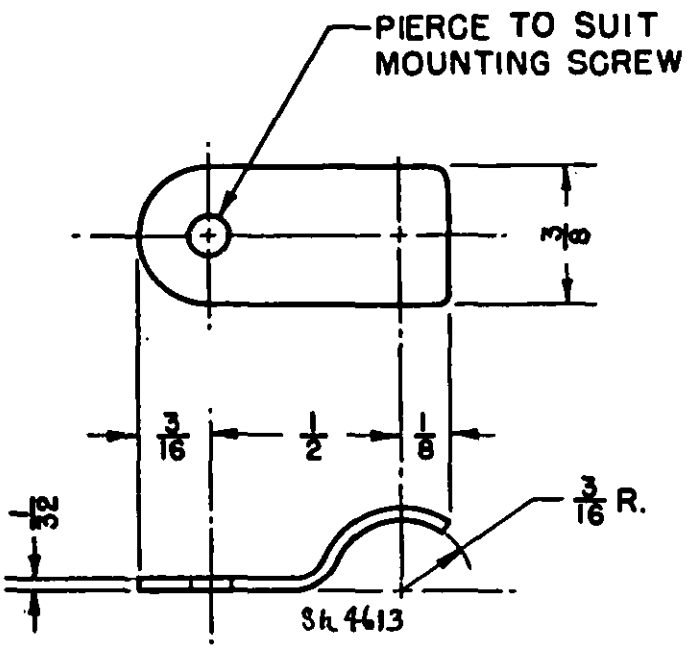
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FIGURE 3. Graduated plates.

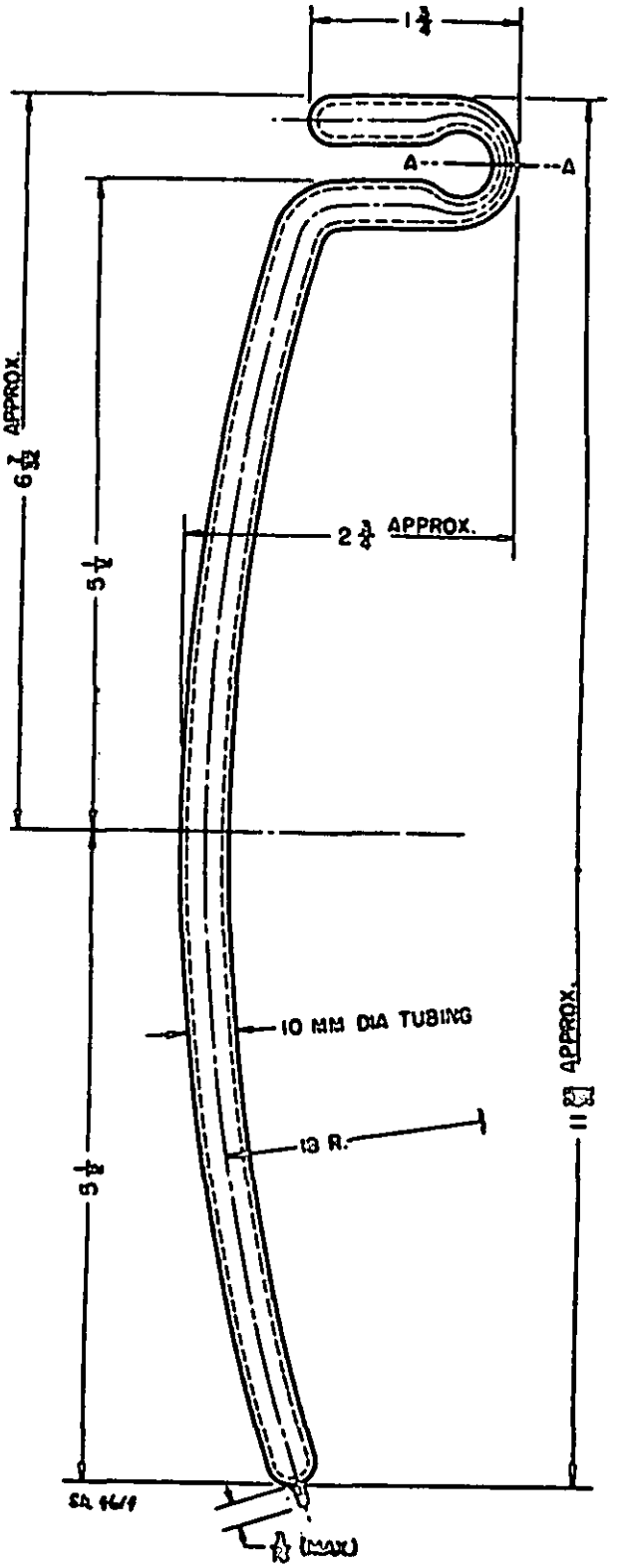
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FIGURE 4. Tube clip.

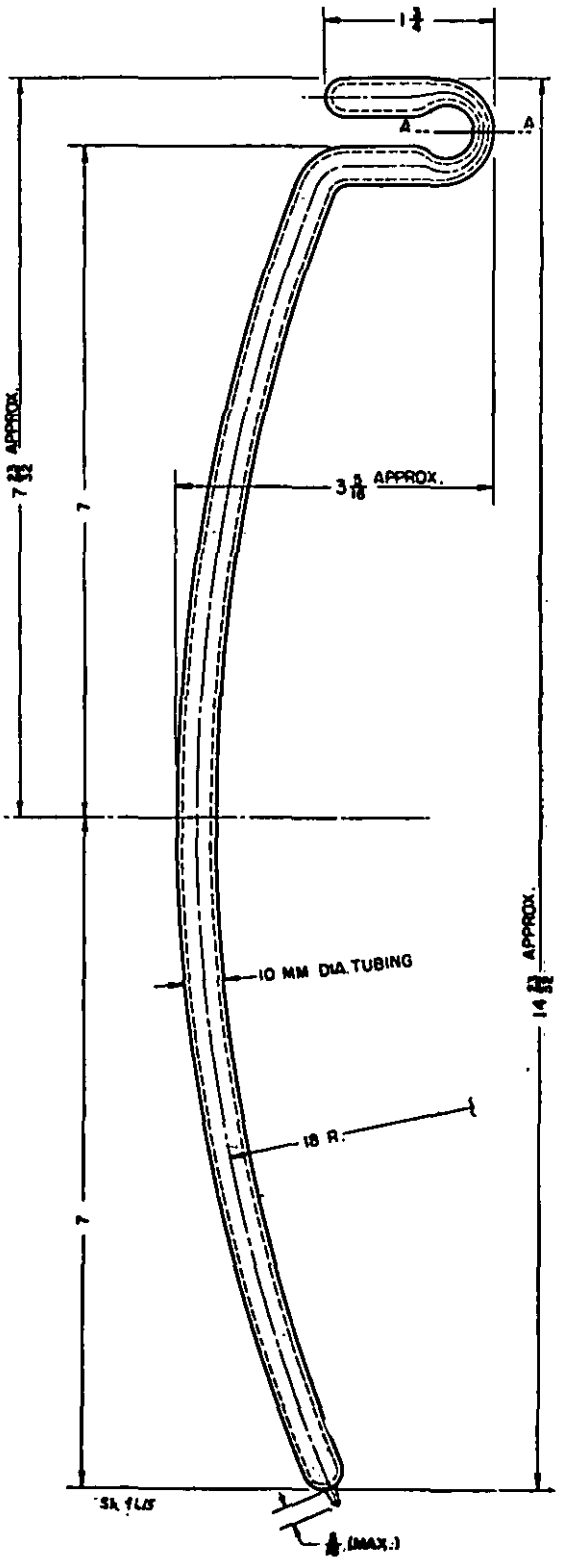
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NOTE: All dimensions are in inches.

FIGURE 5. Type I, class A, 15° (trim indicator).

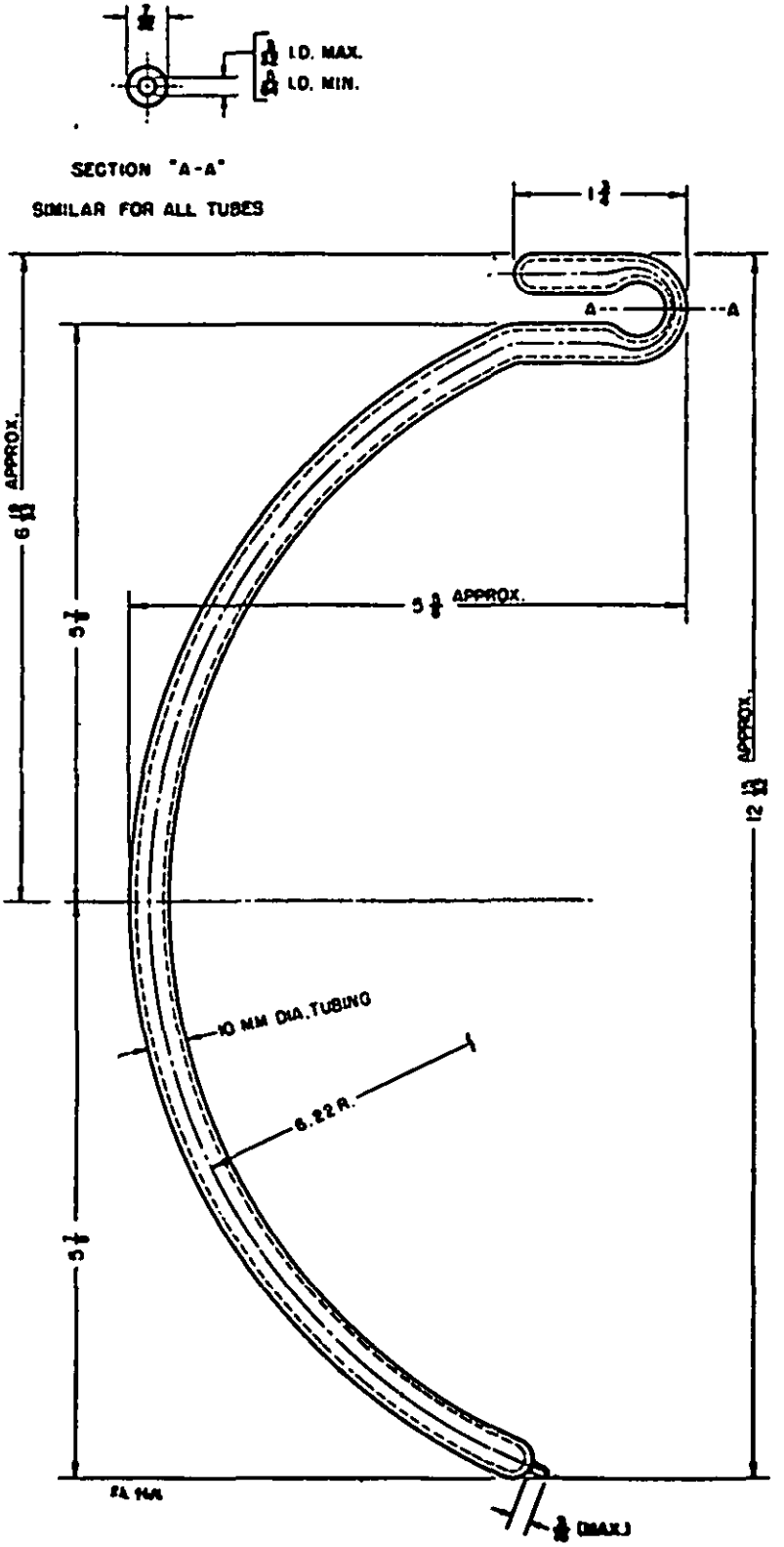
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NOTE: All dimensions are in inches.

FIGURE 6. Type II, class B, 20° (heel indicator).

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FIGURE 7. Type II, class C, 60° (heel indicator).

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